INTRODUCTION

Earlier lessons introduced forward commitment features, payoff profiles, and concepts used in pricing these derivative instruments. In particular, the relationship between spot and forward commitment prices was established as the opportunity cost of owning the underlying asset (represented by the risk-free rate) as well as any additional cost or benefit associated with holding the underlying asset. This price relationship both prevents arbitrage and allows a forward commitment to be replicated using spot market transactions and risk-free borrowing or lending.

In the first lesson, we explore the pricing and valuation of forward commitments on a mark-to-market basis from inception through maturity. This analysis is essential for issuers, investors, and financial intermediaries alike to assess the value of any asset or liability portfolio that includes these instruments. The second lesson addresses forward pricing for the special case of underlying assets with different maturities such as interest rates, credit spreads, and volatility. The prices of these forward commitments across the so-called term structure are an important building block for pricing swaps and related instruments in later lessons.

Learning Module Overview

- A forward commitment price agreed upon at contract inception remains fixed and establishes the basis on which the underlying asset (or cash) will be exchanged in the future versus the spot price at maturity.
- For an underlying asset that does not generate cash flows, the value of a long forward commitment prior to expiration equals the current spot price of the underlying asset minus the present value of the forward price discounted at the risk-free rate. The reverse is true for a short forward commitment. Foreign exchange represents a special case in which the spot versus forward price is a function of the *difference* between risk-free rates across currencies.
- For an underlying asset with additional costs and benefits, the forward contract mark-to-market (MTM) value is adjusted by the sum of the present values of all additional cash flows through maturity.
- Underlying assets with a term structure, such as interest rates, have different rates or prices for different times-to-maturity. These zero or spot and forward rates are derived from coupon bonds and market reference rates and establish the building blocks of interest rate derivatives pricing.
- Implied forward rates represent a breakeven reinvestment rate linking short-dated and long-dated zero-coupon bonds over a specific period.
- A forward rate agreement (FRA) is a contract in which counterparties agree to apply a specific interest rate to a future period.

Learning Module Self-Assessment

These initial questions are intended to help you gauge your current level of understanding of this learning module.

1. At time $t = 0$, the spot price of the underlying asset rises instantaneously and other market parameters remain unchanged.	A. The forward contract buyer has an MTM gain.
2. At time t, the present value of the forward price discounted at the risk-free rate (r) equals the current spot price (S_t).	B. The forward contract seller has an MTM gain.
3. At time T, the forward contract price, $F_0(T)$, is greater than the current spot price, S_T.	C. The MTM value of the forward contract is zero.

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Solution:

1. A is correct. In order to satisfy the no-arbitrage condition, the original spot price, S_0 at $t = 0$, must equal the present value of the forward price discounted at the risk-free rate, r. An immediate increase in the spot price to $S_0^+ > S_0$ results in an MTM gain for the forward buyer.

2. C is correct. At any time t, the MTM value, $V_t(T)$, is equal to the difference between the current spot price, S_t, and the present value of the forward price discounted at the risk-free rate, r, or $F_0(T)(1 + r)^{-(T-t)}$. When $S_t = F_0(T)(1 + r)^{-(T-t)}$, then $V_t(T) = 0$.

3. B is correct. The MTM value to the forward contract seller upon settlement at time T is equal to the settlement value of $F_0(T) - S_T$.

2. An increase in the risk-free rate, r, following the inception of a forward contract will cause which of the following to the forward contract's MTM value to the forward seller if other parameters remain unchanged.

A. The forward contract's MTM value to the forward seller will be unchanged.
B. The forward contract's MTM value to the forward seller will increase.
C. The forward contract's MTM value to the forward seller will decrease.

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Solution:

C is correct. The mark-to-market value from the forward seller's perspective is equal to Vt(T) in the following equation:

$V_t(T) = F_0(T)(1 + r)^{-(T-t)} - S_t$.

An increase in the risk-free rate, r, following the inception of a forward contract will cause the present value of the forward price, F0(T), to fall, and this will reduce the MTM value from the contract seller's perspective.

3. Which of the following is closest to the two-year zero rate given a 3% annual coupon bond priced at 99 per 100 face value if a one-year annual coupon bond from the same issuer has a yield-to-maturity of 2.50%?

A. 3.5266%
B. 3.5000%
C. 3.5422%

Hide Solution

Solution:

C is correct. The yield-to-maturity and the zero rate for a bond with a single cash flow at maturity in one period are identical, so the one-year zero rate, z1, equals 2.50%. Solve for the two-year zero rate (z2) in the following equation:

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Solve for z_2 to get 3.5422%. A is incorrect because 3.5266% is the internal rate of return (IRR) solved for cash flows of –99 at t = 0, 3 at t = 1, and 103 at t = 2. This response assumes a flat term structure, which is not a correct assumption given the question. B is incorrect because this response implies that we can find the correct answer by assuming the coupon rate is the simple average of the one- and two-year zero rates.

4. Which of the following is a correct description of a 2y3y forward rate?

A. The implied two-year rate beginning three years in the future.
B. The implied three-year rate beginning two years in the future.
C. The implied one-year rate beginning two years in the future.

Hide Solution

Solution:

B is correct. In the terminology of forward rates, the first number reflects the point in time when a forward rate begins; thus, the forward rate stated above reflects a rate starting two years in the future. The second number reflects the maturity of the rate. Thus, the 2y3y forward rate reflects a three-year rate starting in two years. A is incorrect because this is the description of the 3y2y forward rate. C is incorrect because this is the description of the 2y1y rate.

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